FARUK AHMED

Prospective Ph.D. Researcher | Medical Computer Vision, Federated Learning & Healthcare AI

a.armaan.faruk@gmail.com +880 1687-301765 Dhaka Cantt-1206, Bangladesh Portfolio LinkedIn GitHub

RESEARCH INTERESTS

- **Medical Image Analysis & CAD:** Deep learning architectures for automated tumor detection (brain and breast), lesion segmentation, and multimodal clinical classification across MRI, CT, Digital Mammography, Ultrasound, and Radiographic X-rays.
- **Federated Learning & Privacy-Preserving AI:** Decentralized optimization across collaborative hospitals under non-IID conditions, modality heterogeneity, divergent label spaces, and communication efficiency (HIPAA/GDPR compliant).
- **Foundation Models & Multimodal Learning:** Adapting vision-language foundation architectures (SAM, Biomedical CLIP) to clinical imaging via Parameter-Efficient Fine-Tuning (PEFT, LoRA, localized adapters).
- **Digital Healthcare & Clinical Informatics:** Cross-modal synthesis of diagnostic imaging with Electronic Health Records (EHR) and clinical retrieval paradigms (RAPTOR) for evidence-backed clinical decision support.

EDUCATION

Southeast University Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering (CSE) 2016 – 2020
CGPA: 3.83 / 4.00

- **Relevant Coursework:** Machine Learning, Design & Analysis of Algorithms, Data Structures, Linear Algebra, Database Management Systems, Software Engineering, Object-Oriented Programming, Discrete Mathematics.

Adamjee Cantonment College Dhaka Cantonment, Bangladesh
Higher Secondary Certificate (HSC) — Science Group 2012 – 2014
Result: GPA 5.00 / 5.00 (Golden A+)

Adarsha Biddya Niketon Manikdi Dhaka, Bangladesh
Secondary School Certificate (SSC) — Science Group 2010 – 2012
Result: GPA 5.00 / 5.00 (A+)

RESEARCH EXPERIENCE & WORKING MANUSCRIPT

Investigating Privacy-Preserving Foundation Model-Based Federated Learning for Healthcare Institutions with Heterogeneous Modalities and Label Sets with a Focus on Communication Efficiency
Faruk Ahmed *(Contributing Researcher)* **(Advanced Stage)**

- Investigating parameter-efficient adapter synchronization (PEFT / LoRA) during decentralized global aggregation to reduce client-server parameter transmission bandwidth overhead by over 70% for resource-constrained clinical settings.
- Formulating multimodal cross-modality representation alignment strategies bridging structural feature divergences across diverse 2D imaging inputs (MRI, CT, digital mammography, ultrasound, radiographic X-rays) without violating institutional data privacy constraints.
- Evaluating decentralized global model robustness against non-IID statistical data skew, inter-hospital acquisition protocol variances, and missing institutional label taxonomies across collaborative healthcare centers.

TECHNICAL COMPETENCIES

AI & Machine Learning:	Medical Image Analysis, Computer Vision, Federated Learning, Foundation Models (SAM, CLIP), Deep Learning, Image Segmentation, Transfer Learning, Health Informatics, Large Language Models (LLMs)
Programming & Data:	Python, C++, SQL, PyTorch, OpenCV, NumPy, Pandas
Quality Engineering:	Test Planning, Test Case Design, Defect Lifecycle Management, Functional/Regression/Smoke Testing, UAT, Cross-Browser/Device Verification, API Testing, Root Cause Analysis
Platforms & Tools:	Git, GitHub, Jira, Postman, VS Code, Google Colab, JMeter, Chrome DevTools
Professional Practices:	Agile/Scrum Methodologies, Technical Communication, International Client Handling, Requirements Analysis, Reproducible Research Protocols

PROFESSIONAL EXPERIENCE

Genie InfoTech Dhaka, Bangladesh
Software Quality Assurance Engineer Sep 2023 – Present

- **Enterprise Platform Verification:** Own end-to-end quality assurance and verification workflows across 3 flagship enterprise platforms: *Limadi* (commercial logistics & fleet routing), *Maway* (role-based EdTech & scheduling), and *Genie Split* (fintech ledger app).
- **Test Engineering & Execution:** Design and maintain comprehensive test suites, Postman API collections, and QA checklists; execute functional, regression, smoke, cross-browser, cross-device, and User Acceptance Testing (UAT).
- **Systems Reliability & Defect Lifecycle:** Isolate complex edge cases and reproducible failure modes using Jira; collaborate closely with development teams on algorithmic verification, backend validation, and release readiness.
- **Stakeholder Leadership:** Participate in sprint planning, foreign client requirements reviews, and technical documentation audits.

Service Engine BPO Dhaka, Bangladesh
Executive — QA & Advertising Operations Jun 2022 – Aug 2023

- Managed quality assurance workflows for Google AdSense campaign trafficking within the Digital Advertising Department.
- Audited targeted digital advertising deployments across thousands of publisher web properties, maintaining 100% compliance with strict formatting, technical parameter specifications, and policy requirements.
- Executed systematic quality control on high-volume trafficking runs, minimizing deployment error rates and enhancing operational throughput.

Education Dot Com Dhaka, Bangladesh
Academic Tutor (Science & ICT) May 2020 – Jun 2022

- Provided curriculum-focused academic instruction in ICT, Chemistry, and science to higher-secondary and secondary students.
- Mentored student for being prepared for their academic board exams with care.

LEADERSHIP & COMMUNITY IMPACT

Southeast University Computer Club Dhaka, Bangladesh
Distinguished Member 2017 – 2020

- Organized competitive programming contests, tech bootcamps, and departmental software engineering workshops.
- Served on the executive management committee for annual departmental galas, academic orientations, and technical exhibitions.

Regional Winter Relief Campaign Northern Bangladesh
Project Lead 2018 – 2019

- Directed regional fundraising campaigns, procurement logistics, and direct distribution of warm winter clothing to over 500 underprivileged families across northern Bangladesh.

Adamjee Cantonment College Dhaka Cantonment, Bangladesh
Band Team Leader — Annual Sports & Ceremonial Events 2013 – 2014

- Led the institutional marching band team during annual athletic ceremonies and state sporting events, coordinating synchronization and team discipline.

PROFESSIONAL TRAINING & CERTIFICATIONS

Software Quality Assurance (SQA) & Cyber Security Jul 2023 – Sep 2023
IT Training BD (US-Aligned Curriculum) Intensive Professional Certification

Digital Ad Operations for Google AdSense Jul 2022
Service Engine BPO (SEBPO), Dhaka, Bangladesh Corporate Operational Certification

ACADEMIC & EXPERIENCE REFEREES

Ashik Mahmud
Team Lead
Software Development
Genie InfoTech
Dhaka, Bangladesh
Email: genie.ashikmahmud@gmail.com

Shahriar Manzoor
Chairman & Associate Professor
Department of Computer Science & Engineering
Southeast University
Room 530, 5th Floor, 251/A Tejgaon I/A, Dhaka-1208
Email: cse.chair@seu.edu.bd, smanzoor@seu.edu.bd